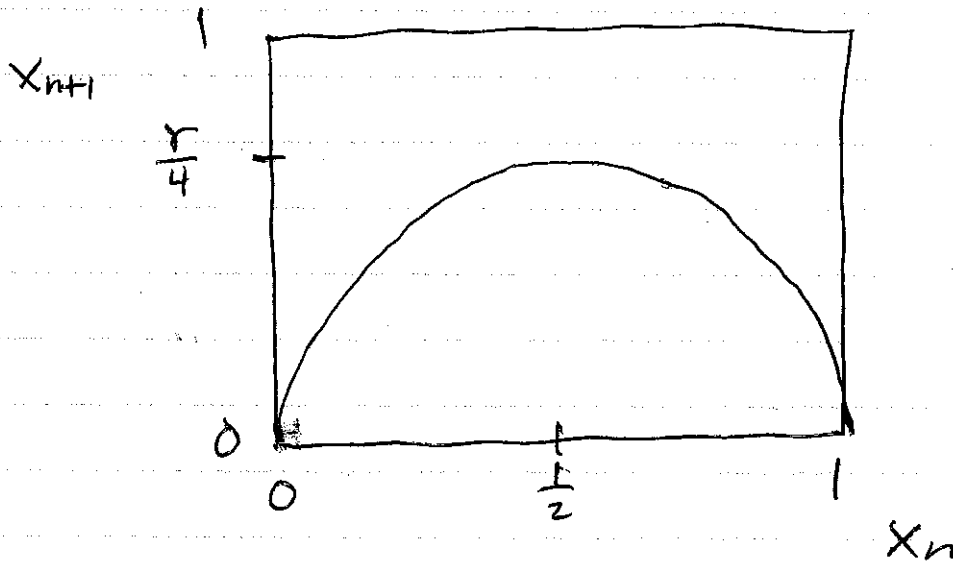


p. 1

Logistic Map

$$X_{n+1} = r X_n (1 - X_n) = f(X_n)$$

$X \in [0, 1] \Rightarrow r \in [0, 4]$ needed so iterates remain in $[0, 1]$

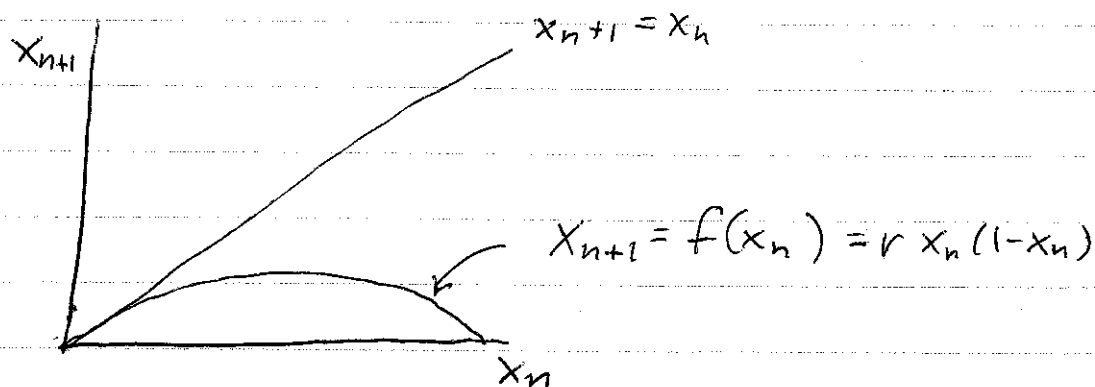


Claims:

- $x^* = 0$ is a fixed pt.
- It is a global attractor for $r \in [0, 1]$
(every initial condition leads to $x_n \rightarrow 0$ as $n \rightarrow \infty$)
- $x^* = 0$ is unstable for $r > 1$
- $x^* = 1 - \frac{1}{r}$ is a fixed pt. for $r > 1$, which is linearly stable for $r < 3$.

$r \leq 1$:

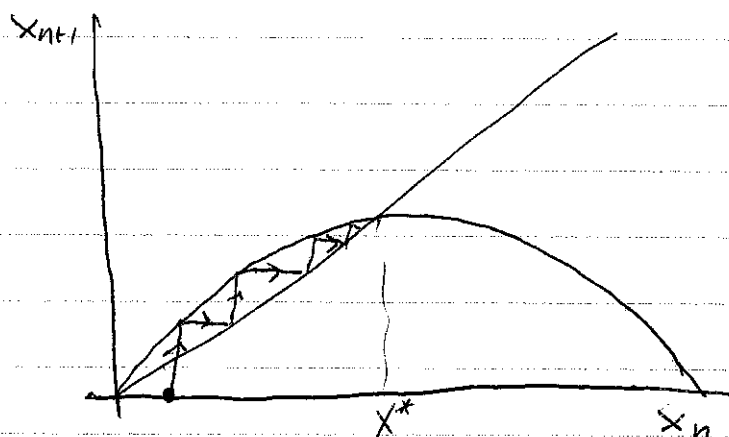
p. 2



$x_{n+1} < x_n$ for $x_n \neq 0$

decreasing sequence converges to $x^* = 0$ ~~point~~

$r > 1$



linear stability: x^* is linearly stable if $|f'(x^*)| < 1$ & unstable if $|f'(x^*)| > 1$

here: $f'(0) = r > 1$

~~fix~~

second pt: $rx^*(1-x^*) = x^* \Rightarrow r(1-x^*) = 1$
 $x^* = \frac{r-1}{r} = 1 - \frac{1}{r}$

p.3

$$\begin{aligned} f'(x^*) &= r(1-x^*) - rx^* = r(1-2x^*) \\ &= r\left(1-2+\frac{2}{r}\right) = 2-r \end{aligned} \quad r > 1$$

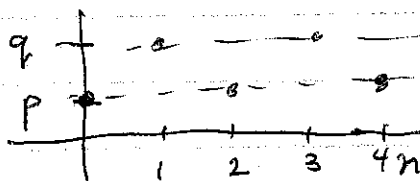
$$|f'(x^*)| = |2-r| < 1 \quad \text{for} \quad r < 3$$

$$f'(x^*) \Big|_{r=3} = -1$$

$r=3$ corresponds to a "period-doubling bifurcation pt" - birth of a period-two cycle occurs at $r=3$:

Two cycle: a pair of points p, q ($p \neq q$) such that $f(p)=q$, $f(q)=p$.

$$p \rightarrow q \rightarrow p \rightarrow q$$



Method to find 2-cycle values p & q :

$$\text{if } f(p)=q \Rightarrow f(f(p)) = f(q)=p$$

so p is a fixed-pt. of $f \circ f(x) = f^2(x)$
& likewise q is a fixed pt of $f^2(x)$

Note that $x=0$ & $x=1-\frac{1}{r}$ are also fixed pts of f^2

p. 4

$$\begin{aligned} f(f(x)) &= f(rx(1-x)) \\ &= r(rx(1-x))(1-rx(1-x)) \\ &= \text{quartic in } x \end{aligned}$$

Fixed-pts satisfy $f^2(x^*) = x^*$ so look for roots of

$$x - f^2(x) = x(1-r+rx)(1+r-\cancel{rx}-r^2x+r^2x^2)$$

$x=0$

$x=1-\frac{1}{r}$

$x=p$
 $x=q$

$$r^2x^2 - (r+r^2)x + (1+r) = 0$$

$$p, q = \frac{(r+r^2) \pm \sqrt{(r+r^2)^2 - 4r^2(1+r)}}{2r^2}$$

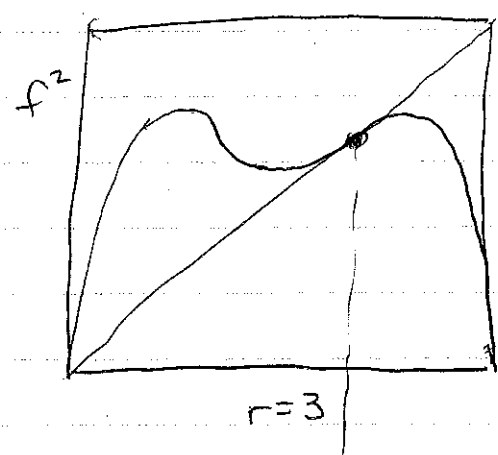
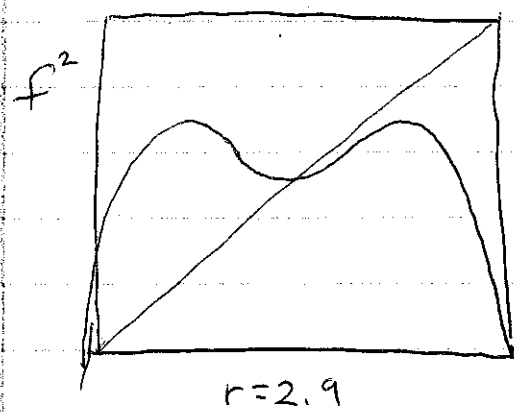
$$= \frac{1+r \pm \sqrt{1+2r+r^2-4-4r}}{2r}$$

$$= \frac{1+r \pm \sqrt{r^2 - 2r - 3}}{2r}$$

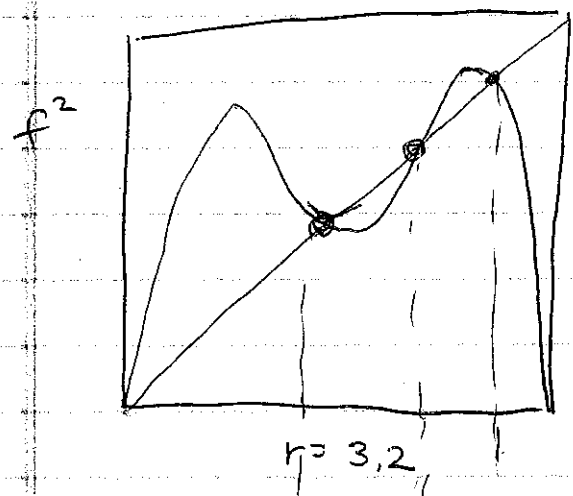
exists for $r^2 - 2r - 3 > 0$
 $(r-3)(r+1) > 0 \Rightarrow r > 3$

$$r \rightarrow 3 \Rightarrow p.g \rightarrow \frac{4}{6} = \frac{2}{3} = 1 - \frac{1}{r} \quad (r=3)$$

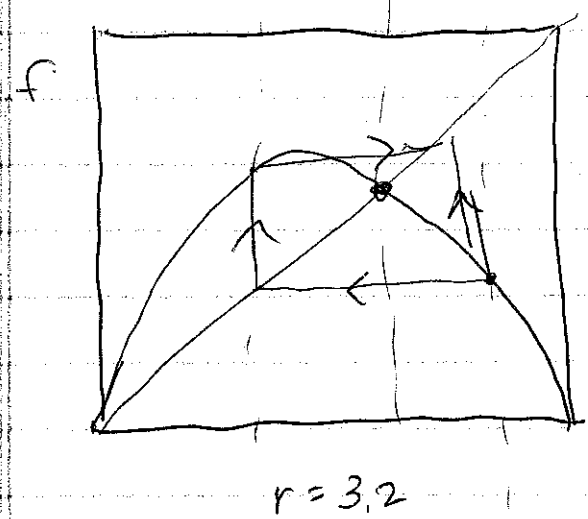
p. 5



$$1 - \frac{1}{r} = \frac{2}{3}$$



$p \quad 1 - \frac{1}{r} \quad q$



Stability of 2-cycle

p & q are both fixed-pts. of the map $f^2(x)$ so we compute stability using it

$$\lambda_p = \left. \frac{d}{dx} [f^2(x)] \right|_p$$

$$\lambda_q = \left. \frac{d}{dx} [f^2(x)] \right|_q$$

) same?

$$\left. \frac{d}{dx} [f(f(x))] \right|_{x=p} = f'(f(p)) f'(p)$$

$$= f'(q) f'(p)$$

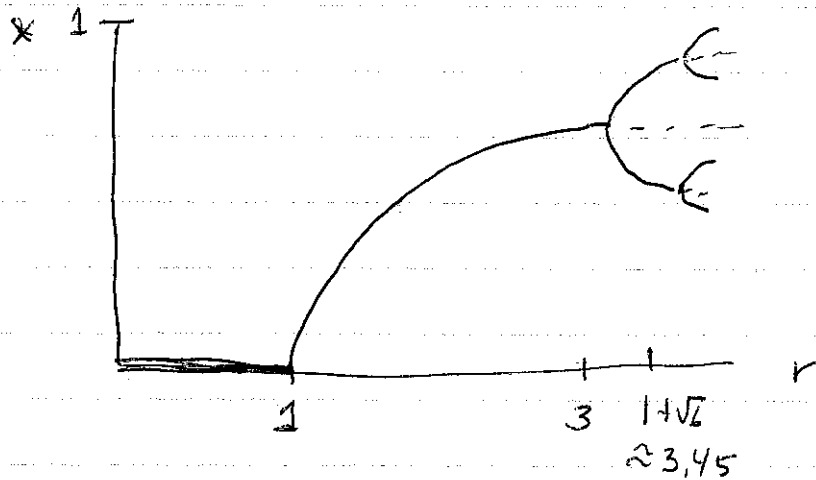
if $|\lambda| < 1 \Rightarrow$ stable 2-cycle

$$\lambda = (r - 2rq)(r - 2rp)$$

$$= r^2 (1 - 2q)(1 - 2p) = 4 + 2r - r^2$$

$$\lambda = -1 \quad @ \quad r = 1 + \sqrt{6}$$

Bifurcation diagram



Logistic Map has a "period-doubling cascade"

$$\underbrace{x^* = 1 - \frac{1}{r}}_{\text{period-1}} \longrightarrow \underbrace{\{p, q\}}_{\text{period-2}} \longrightarrow \underbrace{\{p_1, p_2, q_1, q_2\}}_{\text{period-4}} \longrightarrow \text{period-8} \dots$$

$$r_1 = 3$$

$$r_2 = 1 + \sqrt{6} = 3.449\dots$$

$$r_3 = 3.54409$$

$$r_4 = 3.5644\dots$$

$$r_5 = 3.568759\dots$$

$$r_\infty = 3.569946\dots$$

period 2 born

" 4 "

" 8 "

" 16 "

" 32 "

∞

$r > 3.569946$

& periodic windows are interspersed; period-3 is last one

Orbit diagram construction:

1. Fix r
2. iterate map N times ($N=300$ or so)
& don't record those since they are associated with transient phase
3. iterate map another N times

Plot pairs $(r, x_1), \dots, (r, x_N)$

increment r & repeat

